Model-specific focal points and self-knowledge

GPT-5.4 vs Claude Opus 4.8 · one-shot visual coordination

2026-08-07 · two studies · 6,300 API calls · 0 parse failures, 0 API errors

The two studies

study	question	design	calls
Step 1c screen	which stimuli separate the two models?	27 new pairs × 2 models × n=30	1,620
Step 1c confirmation	which separations survive a fresh sample?	17 stimuli × 2 models × n=60	2,040
Step 2 ground truth	what is the true self-self match?	11 stimuli × 2 models × n=60	1,320
Step 2 self	does a model know its own coordination?	11 stimuli × 2 models × n=30	660
Step 2 auditor	does the OTHER model know it better?	11 stimuli × 2 raters × n=30	660

Headline findings

— Only two stimuli are clean focality disparities on fresh data, and both run GPT > Claude: d_dens_extreme and s_density. There is no clean Claude > GPT disparity in the battery.

— Every GPT > Claude gap is clean — Claude keeps form consistency 0.80–0.97 and merely concentrates less. Every Claude > GPT gap but one has GPT at form consistency 0.43–0.70: GPT failing to find a rule and choosing a side. These are differences in KIND, not in focal strength.

— The winner's curse bit. The screen's only clean disparity, b_spiral_cw, fell from Δ −0.303 to −0.176 on fresh trials; d_eq_ink collapsed from −0.303 to −0.048. At n=30 alone both would have entered the battery.

— Self-knowledge is not symmetric. Claude predicts its own coordination better than GPT does (MAE 0.132 vs 0.176). GPT predicts its own WORSE than Claude does (0.153 vs 0.123). Overall the self-vs-auditor gap is a wash.

— GPT is a near-useless auditor of Claude: its estimates correlate r = −0.07 with Claude's actual coordination. Claude auditing GPT reaches r = +0.55.

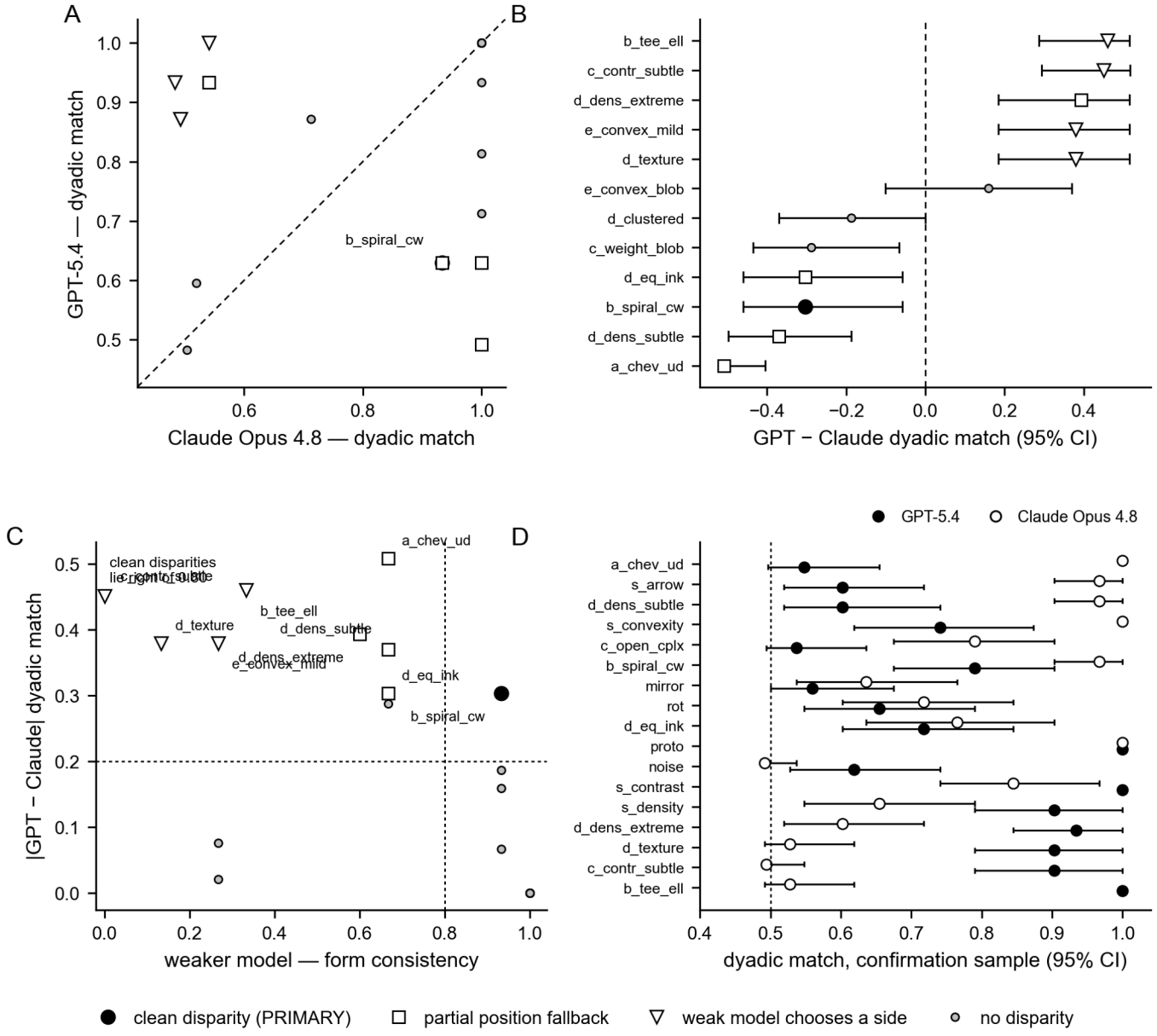
— The self-advantage lives in the position-fallback stimuli, not the clean disparities. On stimuli that merely LOOK coordinable, the focal model knows whether it actually has a rule and the auditor does not.

Procedure held constant across every study

Prompt, model settings and parser are imported from step1_form_choice.py (sha8 3003ae1b), the module Steps 1 and 1b used, so every choice trial in this series shares one estimand. GPT-5.4 reasoning.effort=low; Claude Opus 4.8 max_tokens=2000, adaptive thinking, output_config.effort=low. Neither API exposes a sampling seed; fixed seeds govern AB/BA allocation only. Exact AB/BA counterbalancing in every cell of every stage.

Study 1 — targeted stimulus screen

(A) GPT vs Claude dyadic match, all 27 candidates. (B) Disparity effect sizes, bootstrap 95% CI, 5,000 resamples. (C) The form-consistency diagnostic: clean disparities lie right of 0.80. (D) Confirmation results, fresh n=60 sample.



The position-fallback diagnostic

form_consistency = 1 − | P(choose modal | modal shown LEFT) − P(choose modal | modal shown RIGHT) |

A model choosing by form scores ~1; a model choosing by side scores ~0. Under balanced AB/BA a pure side default reads as ≈0.50 on the identity coding — indistinguishable from indifference on the match rate alone. c_contr_subtle is the extreme case: Claude chose LEFT on 30/30 trials, gave a form-based rationale in 100% of them, and justified opposite choices with opposite rules ("pick the lighter, more salient one" / "pick the darker one as the more salient"). Its 0.483 match rate is not indifference between light and mid-grey; it is a position default wearing a post-hoc form rationale.

Study 1 — confirmed battery

All figures from the fresh n=60 confirmation sample. gFC / cFC = form consistency.

GPT > Claude — clean focality disparity

stimulus	GPT	gFC	Claude	cFC	Δ	95% CI	note
d_dens_extreme	0.934	0.93	0.602	0.80	+0.332	[+0.178, +0.452]	clean
s_density	0.903	0.90	0.655	0.97	+0.249	[+0.080, +0.386]	clean
s_contrast	1.000	1.00	0.845	0.97	+0.155	[+0.033, +0.259]	clean, Δ<0.20

Claude > GPT — one clean, the rest differences in kind

stimulus	GPT	gFC	Claude	cFC	Δ	95% CI	note
b_spiral_cw	0.790	0.90	0.967	0.97	-0.176	[-0.304, -0.031]	**clean, Δ<0.20**
a_chev_ud	0.548	0.67	1.000	1.00	-0.452	[-0.503, -0.345]	GPT partial fallback
s_arrow	0.602	0.60	0.967	0.97	-0.364	[-0.463, -0.226]	GPT partial fallback
s_convexity	0.741	0.70	1.000	1.00	-0.259	[-0.381, -0.127]	GPT partial fallback
c_open_cplx	0.537	0.43	0.790	0.77	-0.253	[-0.392, -0.099]	GPT fallback, FC 0.43

Diagnostics — large Δ, weak model is choosing a side

stimulus	GPT	gFC	Claude	cFC	Δ	95% CI	note
b_tee_ell	1.000	1.00	0.528	0.27	+0.472	[+0.381, +0.508]	NOT a disparity
c_contr_subtle	0.903	0.90	0.494	0.07	+0.410	[+0.294, +0.499]	NOT a disparity
d_texture	0.903	0.97	0.528	0.27	+0.376	[+0.237, +0.475]	NOT a disparity

Controls

stimulus	GPT	gFC	Claude	cFC	Δ	95% CI	note
proto	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	shared-easy positive
rot	0.655	0.97	0.718	0.80	-0.063	[-0.231, +0.115]	intermediate shared
mirror	0.560	0.37	0.636	0.73	-0.076	[-0.217, +0.072]	structured ambiguous
noise	0.619	0.77	0.491	0.13	+0.127	[+0.025, +0.247]	floor

Replication against the earlier screens

stimulus	prior stage	prior Δ	confirmed Δ	replicated
s_arrow	Step 1b	-0.497	-0.364	yes
s_convexity	Step 1b	-0.264	-0.259	yes
c_open_cplx	Step 1b	-0.276	-0.253	yes
s_contrast	Step 1b	+0.370	+0.155	**direction only**
s_density	Step 1b	+0.186	+0.249	yes
b_spiral_cw	screen	-0.303	-0.176	**direction only**
a_chev_ud	screen	-0.508	-0.452	yes
d_dens_extreme	screen	+0.393	+0.332	yes
d_dens_subtle	screen	-0.370	-0.364	yes
b_tee_ell	screen	+0.460	+0.472	yes
c_contr_subtle	screen	+0.451	+0.410	yes
d_texture	screen	+0.379	+0.376	yes
d_eq_ink	screen	-0.303	-0.048	**direction only**

Which families were productive

family	n	median |Δ|	both ≥0.95	PRIMARY
A semantic / canonical orientation	7	0.000	6	0
B canonical abstract glyphs	6	0.076	2	1
C contrast / visual weight	5	0.067	2	0
D density / numerosity / texture	5	0.370	0	0
E convexity / closure	4	0.159	2	0

Study 1 — all 27 screened candidates

n=30 per model. Sorted by |Δ|. Nothing excluded.

stimulus	GPT	gFC	Claude	cFC	Δ	95% CI	screen status
a_chev_ud	0.492	0.67	1.000	1.00	-0.508	[-0.517, -0.405]	SECONDARY
b_tee_ell	1.000	1.00	0.540	0.33	+0.460	[+0.287, +0.515]	POSITION_DEFAULT
c_contr_subtle	0.933	0.93	0.483	0.00	+0.451	[+0.294, +0.517]	POSITION_DEFAULT
d_dens_extreme	0.933	0.93	0.540	0.60	+0.393	[+0.184, +0.515]	SECONDARY
d_texture	0.871	1.00	0.492	0.13	+0.379	[+0.184, +0.515]	POSITION_DEFAULT
e_convex_mild	0.871	1.00	0.492	0.27	+0.379	[+0.184, +0.515]	POSITION_DEFAULT
d_dens_subtle	0.630	0.67	1.000	1.00	-0.370	[-0.497, -0.186]	SECONDARY
b_spiral_cw	0.630	0.93	0.933	0.93	-0.303	[-0.460, -0.058]	PRIMARY
d_eq_ink	0.630	0.67	0.933	0.93	-0.303	[-0.460, -0.058]	SECONDARY
c_weight_blob	0.713	0.67	1.000	1.00	-0.287	[-0.434, -0.067]	REJECT
d_clustered	0.814	0.93	1.000	1.00	-0.186	[-0.370, +0.000]	REJECT
e_convex_blob	0.871	0.87	0.713	0.93	+0.159	[-0.101, +0.370]	REJECT
b_s_z	0.595	0.73	0.519	0.27	+0.076	[-0.103, +0.274]	REJECT
c_fill_outline	0.933	0.93	1.000	1.00	-0.067	[-0.186, +0.000]	REJECT
b_hook_mirror	0.483	0.27	0.503	0.20	-0.021	[-0.145, +0.074]	REJECT
a_arrow_lr	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
a_arrow_ud	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
a_arrow_updiag	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
a_tee_rot90	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
a_tee_upright	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
a_tri_ud	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
b_circ_bar	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
b_cross_x	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
c_contr_extreme	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
c_stroke_wt	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
e_closure_c	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT
e_open_subtle	1.000	1.00	1.000	1.00	+0.000	[+0.000, +0.000]	REJECT

Notes

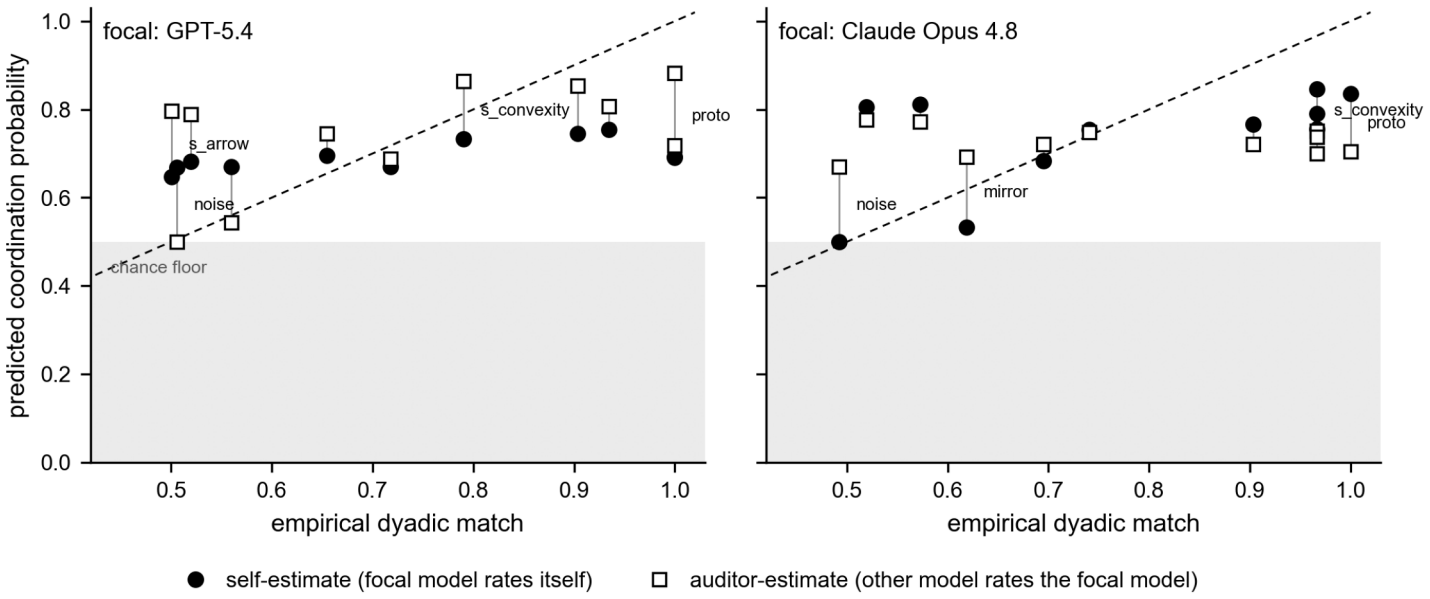
— 12 of 27 candidates sat at 1.000/1.000 for both models. Single-cue manipulations reach ceiling easily; the informative settings are the near-threshold ones.

— Family A collapsed: six of seven canonical-orientation pairs are 1.000/1.000. s_arrow has no sibling that reproduces it, so it is an idiosyncratic GPT failure, not an orientation deficit.

— e_convex_mild (POSITION_DEFAULT, screen Δ +0.379) was cut by the 17-stimulus battery cap and received no confirmation trials. Recorded rather than dropped silently.

Study 2 — self vs auditor calibration

Does a model know when it can coordinate with copies of itself better than the other model does?



Dashed line: perfect calibration. Shaded band: below the 0.50 chance floor for a two-option match.
Vertical segments join the two estimates for one stimulus; their length is the self-vs-auditor gap.

Design

stage	who	n / cell	what is asked
ground truth	each model	60	the actual one-shot choice task
self	each model about itself	30	P(two instances of YOU match?) + which form
auditor	each model about the other	30	P(two instances of the OTHER model match?) + which form

Independence guarantees

— The auditor prompt carries the stimulus image and a third-person restatement of the task and nothing else — no empirical match rate, no counts, no prior model output. Same for self-estimates. Verified by prompt hash: ground truth 3003ae1b (identical to Steps 1/1b/1c); auditor prompts 50aee453 and e23c8d67, one per focal model, with no data embedded.

— Every trial is a fresh single-turn API call with no conversation history.

— AB/BA counterbalancing applies to all three stages. The estimate prompts ask for the likely form as a POSITION in the rater's own image, since the stimuli carry no labels, so balance is required to recode that answer to an identity and stops layout leaking into the probability.

— Ground-truth trials are fresh and never pooled with the Step 1b/1c screen or confirmation data. They reproduce the Step 1c confirmation to within ±0.05 on 8 of 11 stimuli (max |Δ| = 0.11).

Scale. With two options, two independent players match at 0.50 by chance, so the empirical quantity is bounded below at ≈0.50. Estimates below it are miscalibrated by construction and are reported exactly as given, never rescaled. Form consistency is a diagnostic here, never an exclusion rule.

Study 2 — per focal model × stimulus

emp = empirical dyadic match (fresh ground truth). adv = auditor error − self error; positive favours self.

Focal: GPT-5.4 auditor: Claude Opus 4.8

stimulus	role	emp [95% CI]	FC	self	aud	sErr	aErr	adv
s_arrow	fallback	0.501 [0.49, 0.56]	0.40	0.648	0.797	0.147	0.296	+0.149
a_chev_ud	fallback	0.519 [0.49, 0.60]	0.70	0.681	0.788	0.162	0.269	+0.107
rot	control	0.655 [0.55, 0.79]	0.83	0.696	0.745	0.041	0.090	+0.049
s_convexity	fallback	0.790 [0.67, 0.90]	0.77	0.733	0.864	0.057	0.074	+0.016
b_spiral_cw	clean disparity	0.718 [0.60, 0.84]	0.93	0.671	0.688	0.047	0.029	-0.018
s_contrast	clean disparity	1.000 [1.00, 1.00]	1.00	0.693	0.718	0.307	0.282	-0.026
s_density	clean disparity	0.934 [0.84, 1.00]	0.93	0.755	0.808	0.180	0.127	-0.053
mirror	control	0.560 [0.50, 0.67]	0.37	0.671	0.543	0.111	0.017	-0.094
d_dens_extreme	clean disparity	0.903 [0.79, 1.00]	0.90	0.745	0.853	0.158	0.050	-0.108
noise	control	0.506 [0.49, 0.59]	0.57	0.669	0.500	0.163	0.006	-0.158
proto	control	1.000 [1.00, 1.00]	1.00	0.691	0.882	0.309	0.118	-0.191

Focal: Claude Opus 4.8 auditor: GPT-5.4

stimulus	role	emp [95% CI]	FC	self	aud	sErr	aErr	adv
noise	control	0.492 [0.49, 0.54]	0.03	0.500	0.670	0.008	0.178	+0.170
proto	control	1.000 [1.00, 1.00]	1.00	0.835	0.704	0.165	0.296	+0.131
s_convexity	fallback	0.967 [0.90, 1.00]	0.97	0.847	0.751	0.120	0.215	+0.095
a_chev_ud	fallback	0.967 [0.90, 1.00]	0.97	0.790	0.700	0.177	0.266	+0.090
s_arrow	fallback	0.903 [0.79, 1.00]	0.90	0.767	0.721	0.137	0.182	+0.046
b_spiral_cw	clean disparity	0.967 [0.90, 1.00]	0.97	0.756	0.738	0.211	0.229	+0.018
rot	control	0.696 [0.59, 0.84]	0.97	0.683	0.721	0.013	0.026	+0.013
s_contrast	clean disparity	0.741 [0.62, 0.87]	0.97	0.755	0.749	0.014	0.008	-0.006
mirror	control	0.619 [0.53, 0.74]	0.70	0.532	0.693	0.086	0.074	-0.012
d_dens_extreme	clean disparity	0.519 [0.49, 0.60]	0.57	0.805	0.777	0.286	0.258	-0.028
s_density	clean disparity	0.573 [0.51, 0.70]	0.87	0.812	0.772	0.239	0.199	-0.040

Overall calibration

subset	rater	n	MAE	RMSE	r with empirical
ALL	self	22	0.1426	0.1684	+0.5152
ALL	auditor	22	0.1495	0.1810	+0.3499
focal=GPT-5.4	self	11	0.1530	0.1761	+0.6593
focal=GPT-5.4	auditor	11	0.1234	0.1614	+0.5457
focal=Claude Opus 4.8	self	11	0.1322	0.1603	+0.5618
focal=Claude Opus 4.8	auditor	11	0.1756	0.1987	**-0.0729**

Study 2 — conclusions

1. Are self-predictions more accurate than auditor predictions?

Overall, barely — and the answer reverses with which model is focal. Self MAE 0.1426 vs auditor MAE 0.1495 across 22 cells; self is closer in 11 of 22. Underneath that wash, Claude predicts itself BETTER than GPT does (MAE 0.132 vs 0.176) while GPT predicts itself WORSE than Claude does (0.153 vs 0.123).

The sharpest number: GPT's estimates of Claude correlate r = −0.073 with Claude's actual coordination — as an auditor of Claude, GPT carries essentially no signal. Claude auditing GPT reaches r = +0.546. GPT has the highest self-correlation of all four (+0.659) but the worst self-MAE: its ordering is fine, its scale is crushed into a 10-point band (64.8–75.5) against a truth spanning 50 points. Claude's spans 50.0–84.7.

2. Largest self-vs-auditor advantages

focal	stimulus	role	emp	self	auditor	advantage
Claude Opus 4.8	noise	control	0.492	0.500	0.670	+0.170
GPT-5.4	s_arrow	fallback	0.501	0.648	0.797	+0.149
Claude Opus 4.8	proto	control	1.000	0.835	0.704	+0.131
GPT-5.4	a_chev_ud	fallback	0.519	0.681	0.788	+0.107
Claude Opus 4.8	s_convexity	fallback	0.967	0.847	0.751	+0.095
Claude Opus 4.8	a_chev_ud	fallback	0.967	0.790	0.700	+0.090

Three of the top five are cases where the focal model knows it will FAIL and the auditor wrongly expects success. Claude on noise is the extreme: 0.500 on all 30 trials, zero variance, against a truth of 0.492 — while GPT predicts Claude will coordinate at 0.670.

3. Where does the advantage live?

focal	stimulus role	n	self err	auditor err	advantage
GPT-5.4	clean disparity	4	0.1731	0.1220	-0.0512
GPT-5.4	**fallback**	3	0.1222	0.2129	**+0.0907**
GPT-5.4	control	4	0.1559	0.0577	-0.0983
Claude Opus 4.8	clean disparity	4	0.1874	0.1733	-0.0141
Claude Opus 4.8	**fallback**	3	0.1445	0.2214	**+0.0769**
Claude Opus 4.8	control	4	0.0679	0.1434	+0.0755

In the position-fallback stimuli, not the clean disparities — and this holds for BOTH models independently. On clean disparities the self-advantage is negative for both: those stimuli look coordinable and an outside observer reads them correctly, so private access buys nothing. On the fallback family (s_arrow, a_chev_ud, s_convexity) the focal model wins in both directions — GPT knows it flounders where the auditor expects 0.79, Claude knows it succeeds where the auditor expects 0.70. The auditor is anchored to how coordinable the PICTURE looks; the focal model has access to whether it actually has a rule.

Caveats, quality control, and outputs

Caveats

— Both models are compressed toward the middle, so everyone underestimates the high-coordination stimuli (Claude says 0.835 for proto's 1.000; GPT says 0.691). Part of the self-vs-auditor comparison is therefore about which rater is less compressed, not only who is better ordered — which is why MAE and correlation are reported separately.

— The auditor is BETTER at predicting which specific form gets chosen (20/22 vs 18/22). The self-advantage is about knowing one's own reliability, not one's own preference.

— Only one cell met the strict private-advantage bar (empirical ≥ 0.85 and self exceeding auditor by ≥ 0.10): Claude × proto — a shared-easy control, not a disparity. s_convexity (+0.096) and a_chev_ud (+0.090) missed by under a point.

— Claude's ground truth ran ~0.04 below the Step 1c confirmation on 8 of 11 stimuli and above on none. Within per-cell sampling noise, but the consistent sign suggests mild between-run drift. It does not bias the self-vs-auditor comparison, which scores both raters against this same ground truth.

— b_spiral_cw is retained despite a confirmed Δ of 0.176, below the 0.20 target, because it is the only Claude > GPT case where both models track form. Its CI reaches −0.031.

Quality control — all checks passed

stage	trials	valid	cells at intended N	AB/BA exact	dupes	parse fail	API err
1c screen	1,620	1,620	60 / 60	yes	0	0	0
1c confirmation	2,040	2,040	34 / 34	yes	0	0	0
2 ground truth	1,320	1,320	22 / 22	yes	0	0	0
2 self	660	660	22 / 22	yes	0	0	0
2 auditor	660	660	22 / 22	yes	0	0	0

No trials excluded, no stimulus dropped silently, no screen or confirmation data pooled into ground truth, and no candidate selected using confirmation results.

Outputs

directory	files
step1c_output/	targeted_stimulus_manifest.csv
	targeted_screen_calls.csv / _summary / _comparison / _selected
	targeted_confirmation_calls.csv / _summary / _comparison
	targeted_stimulus_screen.png / .pdf
	targeted_stimulus_screen_report.md
	targeted_stimulus_screen_summary.pdf
calibration_output/	calibration_ground_truth_calls.csv
	calibration_self_estimates.csv
	calibration_auditor_estimates.csv
	calibration_summary.csv, calibration_overall.csv
	calibration_self_vs_auditor.png / .pdf
	calibration_report.md

Code: make_targeted_stimuli.py · targeted_common.py · targeted_screen.py · targeted_analysis.py targeted_figure.py · targeted_report.py · calibration_run.py · calibration_analysis.py calibration_figure.py · calibration_report.py · full_report_pdf.py